

Manual

Batch Data Processing TRT-CLA 8.1

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1 Batch Data Processing

Overview

Possible fields of application

The function package "Batch Data Processing" provides functions to collect data that is not part of the "batch" object in relation to batches. This may be manually collected data or data transferred from the order/operation.

The product/material documentation and identification function in production is supported by the print function by which label printing is triggered when batches are posted.

Functions to split and merge batches provide for physical material handling.

Implementation notes

The function package "Batch Data Processing" is used if you would like:

- to define further data for the batch in addition to the information that is provided by default for the "batch" object
- to label and identify produced material (also and in particular WIP material) in the manufacturing process
- to divide existing batches, for logistical reasons, into units that can be identified and handled individually taking traceability into account
- to combine/merge existing batches for logistical reasons taking traceability into account

Integration

The function package refers to batch data of Material and Production Logistics as well as to Tracking & Tracing.

The function packages "Graphic Batch Tracing and Product Documentation" update data to allow for verification of splitting and merging processes of data (traceability).

Functions

- Batch attributes:
 - Collection of batch attributes
 - Capture of additional, industry-specific information on the batch according to configuration
 - Configuration of attributes

- Editing function to define material-specific additional information for the batch including transfer of characteristics from the order and control of display and print performance
- Transfer of batch attributes
 - Automatic transfer of order and operation information as attribute for the batch according to configuration
- Display of batch attributes
 - Presentation of batch attributes in evaluations, reports and overviews
- Printing of batch labels:
 - Trigger for printing when batches are posted
 - Trigger of label printing for special batches or batch-related events at the shop floor terminal
 - Standard label
 - Printing of a standard label with pre-configured layout, printing of batch numbers as plain text and barcode in the Code39 format
 - Individual labels



Individual labels can only be designed if the Designer for labels and shop floor papers has been purchased.

2 Attributes

Summary

Menu	Master data → Material → Attributes
Transaction code	batatt
Function authorization	batatt

Usage

This function is used to create or modify batch attributes in the system.

Integration

The recording function on the terminal can be activated (when changing to a output batch) by assigning batch attributes to a [material type](#).

Requirement

The [material types](#) must already be defined.

Selection criteria

Material type

Only attributes with the selected [material type](#) are selected.

Field index

Only attributes with the selected field index are selected.

Attributes

Only the specified attributes are selected.

When using multiple selection criteria - if nothing else is specified - the amount of overlap of the selection criteria is displayed.

Field descriptions

Material type

The [material type](#) is a key field. The attribute refers to the selection material type (material type = summary of identical materials).

Field index

The field index is a key field. There are 40 text fields, 20 numerical fields and 20 decimal fields from which to select.

Display position

Specifies the order for display on the terminal.

Name

Designation of the attribute.

Unit

Unit of the attribute

Display attribute

If this identifier is set, the attribute is taken into consideration in the display in other masks.

Print attribute on batch ticket, printing position

Reserved; currently no processing.

Capture attribute while generating batch

If this identifier is set, the attribute can be captures during batch generation.

Automatic transfer (additional option)

Field value from the operation or order header can be transferred automatically.
The specification is made for this from:

MES operation or **MES order header** via a field **acronym** that can be selected (e.g. ATK – item/article number, FU:10 - User field 10, etc.)

Data type

Here the data type of the field is specified, including field length and decimal places, if necessary.
However, it is not useful to place a text in a decimal field or decimal input in a text field.

3 Split Batch

Summary

Menu	Material management→ Inventory management→ Batch data overview→ Split batch
Transaction code	batch.split
Function authorization	batch.split

Usage

The split batch function in [batch data overview](#) can be used to split and/or to separate an existing batch into an unrestricted number of batches.

The number of splits will only be restricted by the remaining quantity of batches or in case of merged batches by the remaining subordinate batches/serial numbers. If the sum of the split quantities is higher than the remaining quantity of the original batch, splitting is not possible. It is possible to create splits with yield and/or scrap quantities.

Requirements

A batch can only be split if the following conditions are met:

- the batch must have the status F (free)
- the batch must have a remaining quantity that is higher than 0 or must be assigned to subordinate batches/serial numbers accordingly.
- the user must have the function authorization for splitting

At the moment, it is impossible to:

- split off individual subordinate batches/serial numbers from merged batches that will not be linked up in a new merged batch.
- split scrap batches/locked batches.

Selection criteria

The possible selection criteria are the same as for the [batch data overview](#).

There are two tables to select the respective batches manually.

- Table for viewing all of the selected batches (altogether).
- Table with the batches already selected for splitting.

Field descriptions

The main fields of the function are described in the following:

Splitting "simple" batches:

Quantity to be split off

Entry of the quantity that is to be split off from the existing batch

Batch class

Entry of the batch class to be created

Quantity

Entry of the quantity to be split off

Reason

Entry of the batch reason if the class is scrap

Print report

This indicator must be set to print-out a report stored to the split batch (as an option)

Splitting merged batches:

Batch number

External batch number of the selected merged batch

Internal batch number

System-wide unique batch number of the selected merged batch assigned internally.

Batch class

Batch class of the selected merged batch.

Remaining quantity

Current quantity of the selected merged batch.

Batches to be split off

The display list shows all subordinate batches/serial numbers assigned to the selected merged batch. They are selected and split off accordingly.

Batch class

Batch class to be selected for the new merged batch to be split off.

Reason

Entry of the batch reason if the batch class of the merged batch to be split off is e.g. scrap.

Editing functions (of batch data overview)

In general, the batch data overview provides the following two variants to split batches.

Provided that the function authorization MALUSPLITM is available, the "split batch" function provides two options of splitting/separation. Otherwise, the selection option is disabled and the option "repost remaining quantity of batch to new batch" is set by default.

For "simple" batches:

1. Repost remaining quantity of batch to new batch

If the sum of the split quantities is smaller than the remaining quantity of the batch, this difference quantity will be used to create a new batch (class: yield; status: free). The existing batch will then be set to the "processed" status.

2. Reduce existing batch to remaining quantity

The existing batch will be reduced by the sums of the split quantities. If the remaining quantity of the batch will then be 0, the status will be set to "processed".

For merged batches:

Users select the merged batch they want to split off. If it is a "merged batch" (indicator at the batch), the application shows all subordinate batches/serial numbers assigned to this merged batch.

Users choose (multiple selection) the subordinate batches/serial numbers they want to split off into a new merged batch. Users select the batch class for the new merged batch (and thus for the newly assigned subordinate batches/serial numbers).

After using the "split" function, the new merged batch is posted and the selected subordinate batches/serial numbers are assigned to the new merged batch. To do so, the relevant assignment to the "old" merged batch is deleted. The remaining, unselected sub-batches/serial numbers remain with the existing merged batch (existing number).

4 Merge Batches

Summary

Menu	Material management → Batch management → Batch data overview → Merge batches (new batch number)
Transaction code	batch.merge
Function authorization	batch.merge

Usage

The batch merge function can be used to highlight existing batches using manual selection and merge them into a joint batch. The following "batch types" may be combined:

- "simple" batch and "simple" batch
- Merged batch and merged batch

The following combinations are neither supported nor allowed:

- "simple" batch and merged batch
- Subordinate batch/serial number and merged batch
- Subordinate batch/serial number and subordinate batch/serial number

Selection criteria

The possible selection criteria are the same as for the [batch data overview](#).

There are two tables to select the respective batches manually.

- Table for viewing all of the selected batches (altogether)
- Table with the batches already selected for merging

The tables also contain the typical, current information regarding the individual [batches](#).

Field Descriptions

The most significant fields in the function are the following:

Batch number

The new batch number can be input manually here.

Merge batches (new batch number)

A new batch number is allocated for the merged batches.

Merge batches (use batch number involved)

The batch number of the selected batch is used for all of the merged batches.

Editing functions (of batch data overview)

The "batch data overview" dialog generally provides the following two functions to merge batches. They can be distinguished by the new batch number that is to be used.

**Pool batches (new batch number))**

For "simple" batches:

- The selected batches are merged/pooled. The system will generate automatically a new batch number (which is shown in the dialog) if a new batch number is not entered while executing the function.
- If a new batch number is entered, it will be used. Please note: Batch numbers must be unique within the system!
- The remaining quantity of the new batch is increased by the added quantities of the merged batches. The batch status of the new batch is "free".
- The batches merged in the new batch are set to the batch status "processed" and receive a remaining quantity of 0.

For merged batches:

- The selected merged batches are merged/pooled. The system will generate automatically a new batch number (which is shown in the dialog) if a new batch number is not entered while executing the function.
- If a new batch number is entered, it will be used. Please note: Batch numbers must be unique within the system!
- The remaining quantity of the new merged batch is increased by the assigned subordinate batches/serial numbers. The new merged batch is assigned the status "free".
- The merged batches merged in the new merged batch are set to the batch status "processed" and receive a remaining quantity of 0. The subordinate batches/serial numbers are assigned to the new merged batch.

**Pool batches (use existing batch numbers)**

For "simple" batches:

- The selected batches are merged/pooled. The first batch number of all batch numbers selected in the table view is used for the newly generated batch.

- The remaining quantity of the first or reused batch number is increased by the added quantities of the merged batches. The batch status of the first or reused batch is "free".
- The batches merged in the first or reused batch are set to the batch status "processed" and receive a remaining quantity of 0.

For merged batches:

- The selected merged batches are merged/pooled. The first batch number of all merged batch numbers selected in the table view is used for the newly generated merged batch.
- The remaining quantity of the new merged batch is increased by the assigned subordinate batches/serial numbers. The new merged batch is assigned the status "free".
- The merged batches merged in the new merged batch are set to the batch status "processed" and receive a remaining quantity of 0. The subordinate batches/serial numbers are assigned to the new merged batch.

5 1 Recording Goods Receipt Batch

Overview

Function authorization	batch.generate
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Usage

This function is used to create new batches in the system in the context of recording goods received.

Integration

The batches created can be posted in the system as input batches for plausibility determination and to guarantee traceability of the materials used in the production process.

By recording a new goods receipt batch, the tests for the material received can be generated in quality management.

Requirement

In the basic system settings a prefix has been stored that is used for the automatically generated batch number and specifies the length of the batch number in the system.

The [material types](#) in the system have been updated. If the material type is to be determined automatically by the system, the [assignment between material and material type](#) has been updated.

If the units included with the system upon delivery are not sufficient, you have defined your own [units](#) in the system.

Field descriptions

Workplace

The workplace used for receiving can be recorded for traceability.

MES order number

An order / operation can be recorded for traceability.

Material

Mandatory field for the material number of the batch to be created

Material type

Material type of the batch to be created.

Batch class

Specification regarding the quality of the goods receipt batch.

Yield – The batch is created as yield with the batch status "Approved".

Scrap – The batch is created as yield with the batch status "Blocked".

Quantity

Quantity received

Unit

Unit with regard to quantity

Material buffer

Material buffer into which the batch is received

Transport unit

[Transport unit](#) used for the batch

Comment

Free comment text

Badge

Badge number of the person responsible for recording

6 1 Repost

Summary

Menu	Material management → Inventory management → Batch data overview → Repost
Transaction code	batch.relocate
Function authorization	batch.relocate

Usage

The repost function can be used to repost a batch in another material buffer.

Field descriptions

Batch

Batch number to be reposted.

Workplace

Workplace at which the reposting is carried out

MES order number

MES order number of the batch to be reposted

Material

Material number of the batch to be reposted

Material type

Material type of the batch to be reposted

Material buffer

[Material buffer](#) into which the batch will be reposted

Reason

The relevant reason is entered here based on the batch class (normally, a scrap reason). The reason is kept when reposting.

Comment

Free comment text

Batch class

Class of the batch (normally, yield/scrap)

Badge number

Badge number of the person that performs the reposting

7 Applications provided in MOC:

Overview

Application	Documentation	Transaction code	Function authorization
Attributes	MOC_BatchAttributes.pdf	batatt	batatt
Split batch	MOC_SplitMESBatch.pdf	batch.split	batch.split
Merge batch	MOC_MergeMESBatch.pdf	batch.merge	batch.merge
Generate batch	MOC_BatchGenerate.pdf	batch.generate	batch.generate
Relocate/repost batch	MOC_RelocateMESBatch.pdf	batch.relocate	batch.relocate